

Software Requirements Engineering

Assignment-4: Establishing Project Scope and Requirements Baseline - (CLO-3)

Group Information:

Group Number: 19

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1. Project Scope

Scope Elements:

Product Functionality:

Key Features And Functionalities:

- Online coordination between Railway departments
- Real-time updates for both passenger and employee sides.
- Secure access of user data to only authorized personnel
- Web-based from anywhere, for both passengers and employees

Project Resources:

- **Team Size:**
 - 14 people
- **Skill Set:**
 - 1 Project Owner (Stakeholder)
 - 1 Project Manager
 - 3 Software Developers
 - 2 UI designers
 - 2 Quality assurance engineers
 - 3 Requirement engineers
 - 1 System architect
 - 1 Scrum master
- **Tools available:**
 - **IDE:** VS Code (HTML, CSS & Javascript, Node.js, React.js)
 - **Database:** MongoDB
 - **Task Management:** Scrumwise
 - **Requirements Management:** Pivotal Tracker
 - **Software Life Cycle Management:** Code beamer ALM

Available Time:

→ Total Time: 5 Months:

- ◆ **Requirements analysis:** 2 Months
- ◆ **System design:** 1 Month

- ◆ **Implementation:** 1 Months
- ◆ **Testing:** 29 Days
- ◆ **Deployment:** 1 Day

Brook's Law:

If we have to add additional labor or resources to the ongoing project, we'll use Brook's law, which states that ***"adding labor to a late software project makes it even later."***

Following this law, we will not add new people to the project, rather we will

- **Reevaluate and Prioritize Tasks:**
 - We will first identify the critical tasks to complete.
- **Remove bottlenecks:**
 - Look for bottlenecks, and remove them for a more efficient workflow.
- **Reduce Scope:**
 - Arrange meetings with stakeholders to reduce the scope and delay non-critical features to be released in the future.

2. Analyze Scope Feasibility

Available Timeline: 5 Months

Available Team Members: 14 Team members

Features To Work On:

- Online coordination between Railway departments
- Real-time updates for both passenger and employee sides.
- Secure access of user data to only authorized personnel
- Web-based from anywhere, for both passengers and employees

Total Number Of Features: 4

Evaluation of effort vs available resources & time:

The average effort required per feature: 2 Months

Total time required according to this evaluation: 8 Months

Available Time: 5 Months

Exceeding Time: 3 Months

Project is achievable or over-scoped: The estimated time for completion of the project is 8 months whereas the available time is 5 months, therefore, the project is *over-scoped*

3. Project Scope Reduction

Project Features Before Reduction:-

1. Online coordination between Railway departments.
2. Real-time updates for both passenger and employee sides.
3. Secure access of user data to every Employee.
4. Web-based from anywhere, for both passengers and employees.

Reduction in Feature 1:

There will be no reduction in this feature as our main objective is to make the coordination between departments as smooth and quick as possible so this feature must be implemented as a whole otherwise there is no point in delivering the project which is unable to satisfy the solution of a root cause.

Reduction in Feature 2:

In this feature, the real-time updates will only be delivered to the employees but not the customers as developing the functionality of customers will take time and our main objective was to resolve the issue of coordination between departments so we will prioritize the functionality of railway employees over customers.

Reduction in Feature 3:

In this feature the data access of employees (that includes their reports, their performance, their salary information, and other details) will only be available to the top manager rather than middle managers and front line and, they will only be able to submit the data but not be able to view it as a whole and do proper tracking. The proper access to the data of employees will only be given to top managers. As the data access will be given to employees in a hierarchical order and each employee will have a different interface to view and track the data and additional requirements to handle any changes the manager who is not a top manager will make, so implementing the functionality of data access will have very big module and we can reduce it to top managers and also when top manager will have all the data he can easily consult with the front line and middle managers through the main communication channel provided through **feature 1**.

Reduction in Feature 4:

In this feature our application will only be accessible to employees and not to customers as we have discussed in the reduction of **feature 2** we are not going to implement the functionality of customers as that can lead to the unfulfillment of the solution of the root problem as it will be a bigger module itself and will require more time.

Conclusion:

Final features after scope reduction:

1. Online coordination between Railway departments.
2. Real-time updates for only employee sides.
3. Secure access of user data to only top managers.
4. Web-based from anywhere, for only employees.

These features can be successfully implemented in 5 months on the agreement of the stakeholders.

4. Establish Requirements Baseline

Functional Requirements:

The system should solve the problem of "Delay in train departure and arrival". -----Must Have

The system should maximize coordination between railway departments through an online website. -----Must Have

The user data should be visible to all employees -----Could Have

There should be real-time engine Monitoring -----Must Have

The Passenger is unable to buy Tickets online and Check status of his Booking. -----Won't Have

Non Functional Requirements:

The system user age is from children to adult. -----Must Have

The system should comply with legal constraints from the Ministry of Railways. -----Must Have

The system should be highly available and reliable. -----Should Have

Scalability for future expansions (such as adding Passenger Module for Buying tickets Online and Check his booking Status). -----Could Have
